The education-obesity link across countries over time

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Part 1 - Identify a Social Problem

The social problem we have identified is the link between the BMI Index and educational attainment level, bridging two data sets from Eurostat to compare and identify any correlation between the two factors.

1.1 Describe the Social Problem

The relevance of these two variables, BMI and education levels, and how they are correlated with each other lies in the fact that obesity comes with serious health risks, such as diabetes and cardiovascular issues, which are often more common among people with lower educational attainment (World Health Organization, 2022; European Commission, 2024). This then results in the burden of health risks being disproportionately put on the lower educated class of society, and further increases social inequality issues. Beyond the health risks themselves, education has a clear impact on the ability of a person to maintain a healthy lifestyle, as higher education often comes with higher income, better access to higher quality food and so on. This means that addressing obesity cannot be done through increasing awareness alone, and that broader structural interventions, such as food pricing policies or workplace conditions need to be considered.

The fact we are looking at the European aspect of this issue also adds to its relevance. By looking at multiple countries over time, it becomes possible to assess whether this education gap in obesity rates is a universal pattern across Europe or whether certain countries have managed to narrow it. This kind of cross country comparison is useful for understanding what policy approaches may actually work in practice, especially since educational inequalities in obesity have been documented across Europe in recent research (World Health Organization, 2022; European Commission, 2024).

While the existence of an educational gradient in obesity is well established in the literature (World Health Organization, 2022; European Commission, 2024), far less is settled about how this gap behaves over time and whether it is narrowing or widening across different European countries. Much existing work documents the association at a single point in time or within individual countries, which leaves the question of long-term, cross country trends comparatively underexplored. This is the gap our project addresses: by tracking obesity rates by education level across multiple European countries and survey years, we aim to establish whether the education obesity divide is a stable, universal pattern or one that some countries have managed to reduce.

Part 2 - Data Sourcing

2.1 Load in the data

Both datasets are published by Eurostat. We downloaded them once and saved them as CSV files in the `data/` folder of our repository, which we read into R with `read_csv()`. Using saved copies rather than querying Eurostat live means the report always knits from the same data and does not depend on a network connection or on the Eurostat servers being reachable. The files are small, so storing them in the repository keeps the project self-contained and reproducible: anyone who clones it can knit the report without extra setup.

```r
# Load the saved datasets from our repository
bmi_edu <- read_csv("data/bmi_edu.csv")
obesity_trend <- read_csv("data/obesity_trend.csv")
```

2.2 Provide a short summary of the dataset(s)

```
head(bmi_edu, 3)
#> # A tibble: 3 x 9
#>   freq  unit  bmi        isced11 sex   age   geo   TIME_PERIOD values
#>   <chr> <chr> <chr>      <chr>   <chr> <chr> <chr>       <dbl>  <dbl>
#> 1 A     PC    BMI18P5-24 ED0-2   F     TOTAL AT           2014   45.6
#> 2 A     PC    BMI18P5-24 ED0-2   F     TOTAL AT           2019   40.9
#> 3 A     PC    BMI18P5-24 ED0-2   F     TOTAL BE           2014   45.7
head(obesity_trend, 3)
#> # A tibble: 3 x 6
#>   freq  unit  bmi      geo   TIME_PERIOD values
#>   <chr> <chr> <chr>    <chr>       <dbl>  <dbl>
#> 1 A     PC    BMI25-29 AL           2022   44.8
#> 2 A     PC    BMI25-29 AT           2008   36.5
#> 3 A     PC    BMI25-29 AT           2014   33.3
```

Our first dataset, bmi_edu (Eurostat code hlth_ehis_bm1e, Body mass index (BMI) by sex, age and educational attainment level), comes from the European Health Interview Survey (EHIS), in which respondents report their own height and weight (Eurostat, n.d.-a). For each country and year it gives the share of people in each BMI category, broken down by education level, sex, and age. EHIS runs only every few years, so the data covers mainly the 2014 and 2019 waves. This is the dataset that links weight to education, so it is central to our analysis.

Our second dataset, obesity_trend (Eurostat code sdg_02_10, Obesity rate by body mass index (BMI)), is one of the EU's Sustainable Development Goal indicators, also based on self-reported health surveys (Eurostat, n.d.-b). It does not split the figures by education, but it covers many more years (2008, 2014, 2017, 2019, 2022, 2025), which makes it well suited to tracking obesity over time. It supplies the "across countries, over time" side of our research question. The data are reported by country, covering the EU member states together with several additional European countries, each shown as a two-letter code in the geo column.

2.3 Describe the type of variables included

Table 1: Variables in the two datasets.

Variable	Type	Dataset	Meaning
freq	character	both	data frequency (A = annual)
unit	character	both	unit of measurement (PC = percentage)
bmi	character	both	BMI category (e.g. overweight, obese)
isced11	character	bmi_edu	education level (ISCED 2011 group)
sex	character	bmi_edu	sex (M, F, T = total)
age	character	bmi_edu	age group
geo	character	both	country (Eurostat code)
TIME_PERIOD	numeric	both	survey year
values	numeric	both	share of people in that category

As Table 1 shows, the variables fall into three groups. The health measure is BMI/obesity itself, the share of people in each weight category (the `bmi` and `values` columns). The socioeconomic measure is education (the `isced11` column in `bmi_edu`), which is the variable at the heart of our question. The remaining columns (`sex`, `age`, `geo` for country, and `TIME_PERIOD` for year) are background characteristics that we use to split the figures by group, place, and time.

Neither dataset comes from administrative records; both rely on surveys in which people report their own height and weight. This has two implications. First, self reported data tends to underestimate obesity slightly, since people often round their weight down. Second, the two datasets are not fully independent: the SDG indicator is partly built on the same EHIS survey as `bmi_edu`, so they are related rather than separate sources. We combined them deliberately: `bmi_edu` provides the education breakdown, while `obesity_trend` adds the longer time span. Because both are self reported survey data, our analysis can describe associations (for example, that obesity is more common among lower educated groups) but cannot show that education causes obesity.

Part 3 - Quantifying

3.1 Data cleaning

Data is cleaned in three steps.

Cleaning Step 1: Filter and clean the first dataset

```
bmi_edu_clean <- bmi_edu %>%
  filter(
    bmi == "BMI_GE30",                        # Keep only Obese rows
    isced11 %in% c("ED0-2", "ED3_4", "ED5-8"), # Separate the 3 education groups
    sex == "T",                              # Combined sexes
    age == "TOTAL"                          # Combined adult age group
  ) %>%
  select(geo, TIME_PERIOD, isced11, bmi_value = values) %>%
  drop_na(bmi_value)
```

Step 1: We filter out the bmi_edu dataset, to isolate obesity rates(BMI_GE=>30) across three core education levels (ED0-2 for low, ED3_4 for medium and ED5-8 for high). We restricted the demographics to combined sexes (T) and combined adult age group (TOTAL) to establish a baseline for comparing education levels against obesity prevalance.

Cleaning Step 2: Filter and clean the second dataset

```
obesity_trend_clean <- obesity_trend %>%
  filter(
    bmi == "BMI_GE30" # Keep only Obese rows since that's what we are comparing
  ) %>%
  select(geo, TIME_PERIOD, trend_value = values) %>%
  drop_na(trend_value)
```

Step 2: We filter out the obesity_trend dataset to keep only the obesity figures (BMI>=30).This isolates the specific comparative baseline needed to validate against our education-stratified data.

Cleaning Step 3: Combine both datasets into one master file

```
combined_dataset <- inner_join(bmi_edu_clean, obesity_trend_clean, by = c("geo", "TIME_PERIOD"))
```

Step 3: We perform a multi-key inner_join to merge bmi_edu_clean and obesity_trend_clean by country(geo) and year (TIME_PERIOD).This aligns the education-specific obesity rates side-by-side with national obesity trends for comparison by country and year.

These cleaning choices involve trade offs worth flagging. The inner_join keeps only the country year combinations present in both datasets; because bmi_edu is collected only in 2014 and 2019, the merged combined_dataset is effectively limited to those two waves, even though obesity_trend on its own runs from 2008 to 2025. This is a deliberate cost: the merge buys us the education breakdown but sacrifices time depth, which is exactly why the temporal and event analyses below draw directly on the unmerged obesity_trend_clean instead. Filtering to combined sexes and the total age group, and dropping rows with missing values, likewise trades finer grained coverage for a clean, comparable baseline. A natural refinement would be to keep the two cleaned datasets separate and merge them only where an analysis genuinely needs both, rather than treating the joined file as the single source for everything.

3.2 Generate necessary variables

We created three new variables which add analytical value

Variable 1

```
# Create Variable 1: Human-readable education levels
bmi_edu_with_var1 <- combined_dataset %>%
  mutate(
    edu_level = case_when(
      isced11 == "ED0-2" ~ "Low Education",
      isced11 == "ED3_4" ~ "Medium Education",
      isced11 == "ED5-8" ~ "High Education"
    )
  )
```

The variable `bmi_edu_with_var1` transforms ISCED codes into human readable education levels, by mapping the codes to an explicit categorical variable (edu_level).

```
# Reshaping before subtracting
gap_data <- bmi_edu_with_var1 %>%
  # 1. Isolate the columns we need
  select(geo, TIME_PERIOD, edu_level, bmi_value) %>%

  # 2.Flip the rows into columns
  pivot_wider(
    names_from = edu_level,
    values_from = bmi_value
  )
```

To calculate the gap in obesity between different education levels by country and year, the data must be shaped from a long to a wide format using pivot_wider(). This transforms the values of edu_level into distinct columns (Low Education, Medium Education, High Education), positioning the respective obesity rates for a single country and year onto a single row.

Variable 2

```
# Create Variable 2: Calculate the obesity inequality gap
final_gap_data <- gap_data %>%
```

```r
  mutate(
    obesity_gap = `Low Education` - `High Education`
  )
```

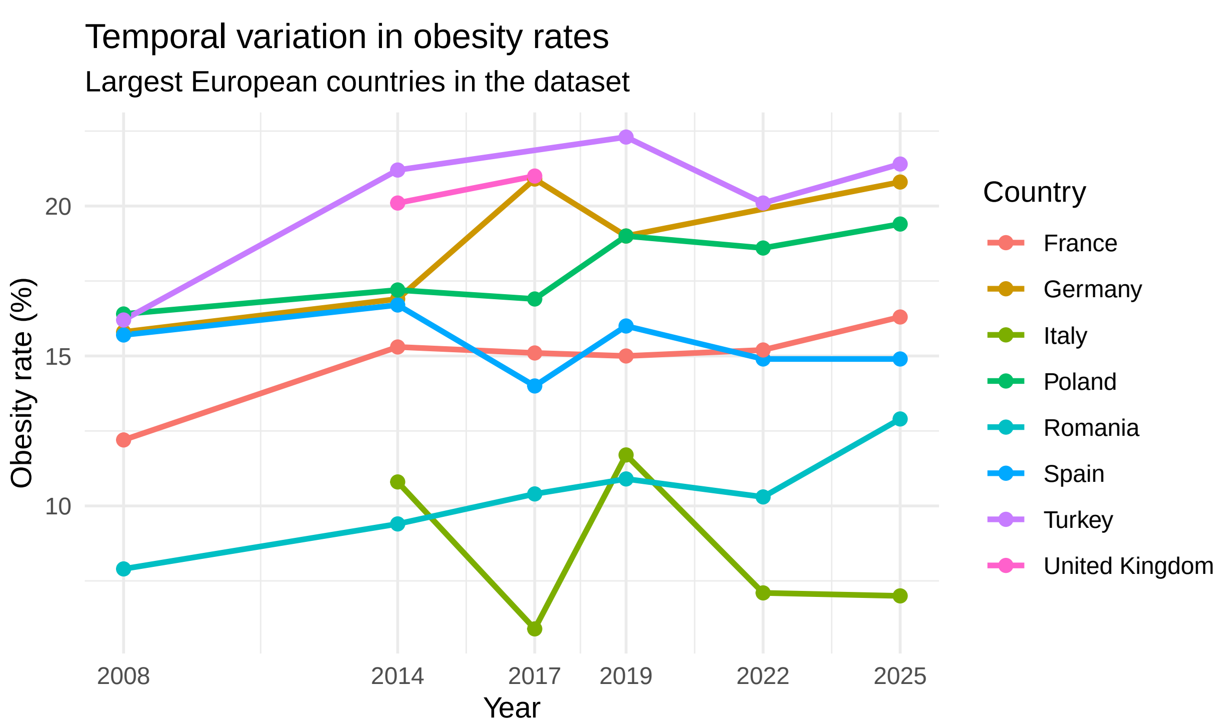
The variable final__gap__data calculates the gap in obesity between education levels by subtracting the obesity rate of the highest education group from the lowest education group. A positive obesity__gap value indicates that obesity is more prevalent among individuals with lower education. This metric allows us to directly evaluate and compare the health inequalities across different European nations and trends over time.

Variable 3

```r
# Create Variable 3: How did the gap change between 2014 and 2019?
gap_change_data <- final_gap_data %>%
  select(geo, TIME_PERIOD, obesity_gap) %>%
  pivot_wider(
    names_from = TIME_PERIOD,
    values_from = obesity_gap,
    names_prefix = "gap_"
  ) %>%
  mutate(gap_change = gap_2019 - gap_2014) %>%
  drop_na(gap_change)
```

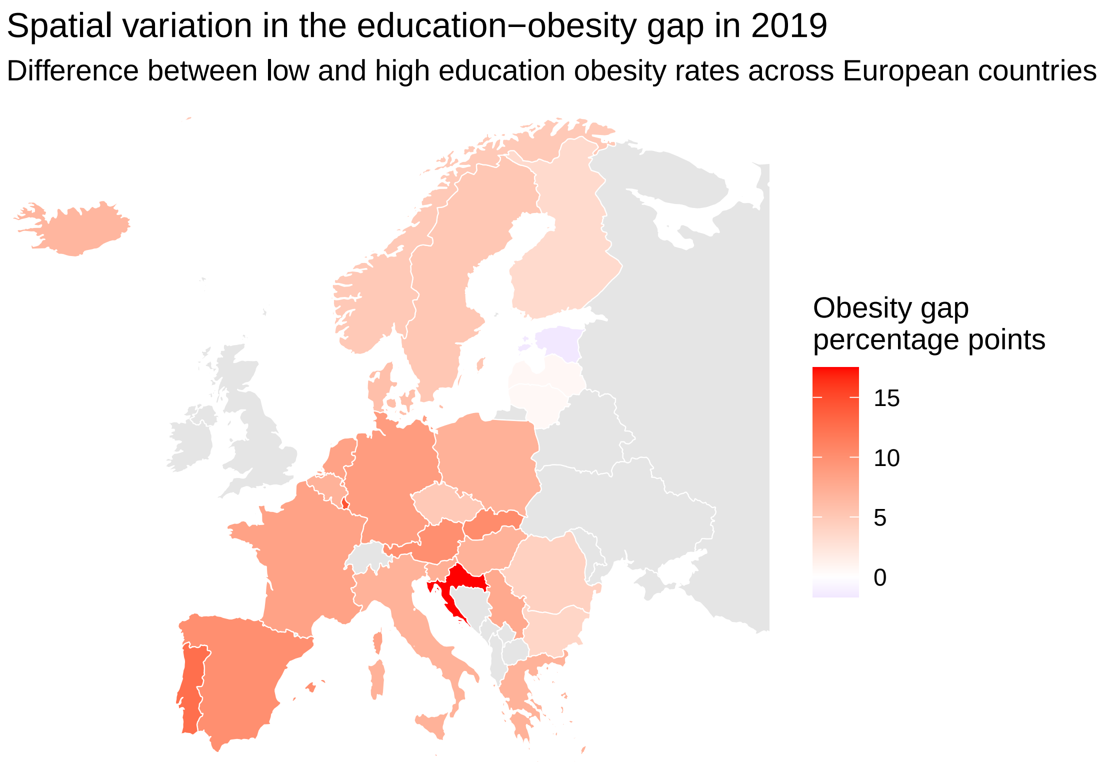
The variable `gap_change` measures whether educational inequality in obesity widened or narrowed in each country between the two survey waves. A positive value means the gap between low and high educated groups grew; a negative value means the country managed to reduce it. This goes directly to the core of our research question, since it separates countries where the problem is getting worse from countries that may hold policy lessons.

3.3 Visualize temporal variation



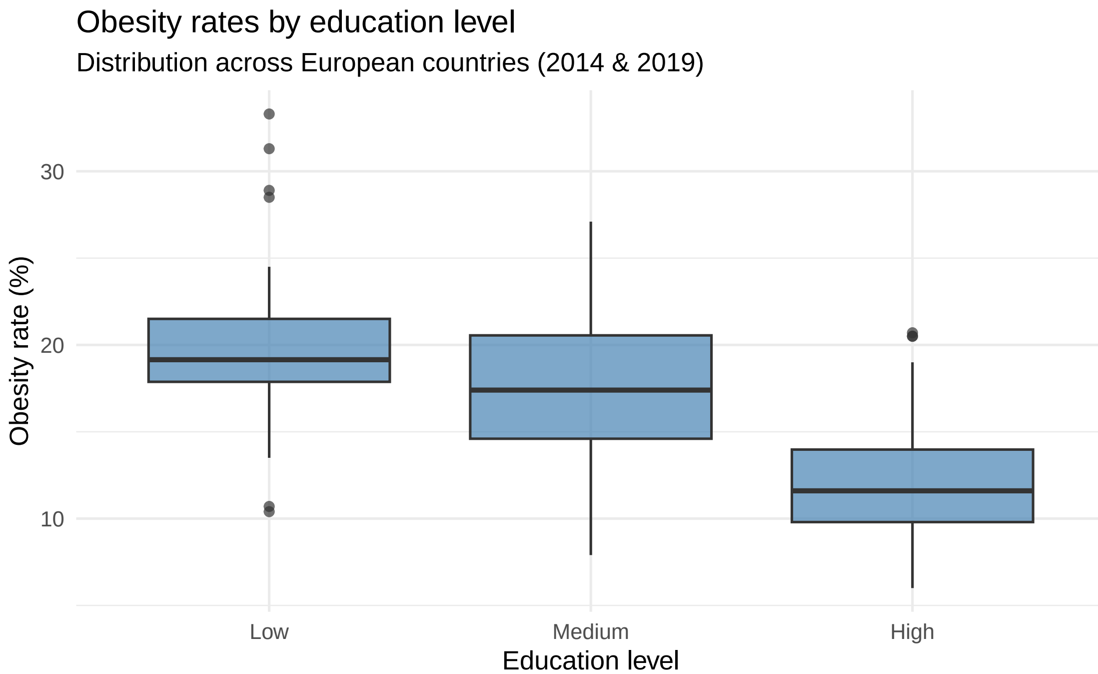
This figure shows temporal variation in obesity rates for the largest European countries available in the dataset. Full country names are used in the legend to make the graph easier to interpret. Across the full period the general direction is upward: most of these countries show higher obesity rates in their latest year than in their earliest, consistent with a broad rise in obesity across Europe. The levels, however, stay clearly separated. Turkey and the United Kingdom sit near the top throughout, while Romania and Italy remain the lowest, and the gap between the highest and lowest prevalence countries does not noticeably close over time. A few series are also non monotonic, Italy in particular swings up and down between waves, which likely reflects the sparse, every-few-years nature of the survey rather than genuine year to year reversals. Some country lines start later than others because not all countries have data available for every year in the dataset. When a country has missing observations in earlier years, the line only starts from the first year for which data is available. Therefore, differences in starting points reflect differences in data availability, not necessarily differences in when obesity became relevant in those countries.

3.4 Visualize spatial variation



This map shows spatial variation in the education-obesity gap across European countries in the latest available year. The education-obesity gap is calculated as the obesity rate among people with low education minus the obesity rate among people with high education. A higher positive value means that obesity is more common among the low educated group compared with the high educated group. The gap is far from uniform across Europe. It is widest by some margin in Croatia, which stands out as the clear outlier, with other comparatively large gaps in southern and south eastern Europe. At the other end, the gap is small in much of the north, and Estonia is a notable anomaly where the pattern is close to zero or even reversed, high educated groups are about as likely to be obese as low educated ones. Read this way, the map shows that educational inequality in obesity is not a single European pattern but is concentrated in particular regions rather than spread evenly across the continent. A map is used because the goal is to visualise differences across geographic space. By placing the values directly on a map of Europe, it becomes easier to see whether the education-obesity gap is larger or smaller in specific parts of Europe. Countries shown in grey either have no available data for the selected year or could not be matched correctly with the map data. To reduce matching problems, country codes from the dataset were converted into full country names before joining the obesity data to the map.

3.5 Visualize sub-population variation



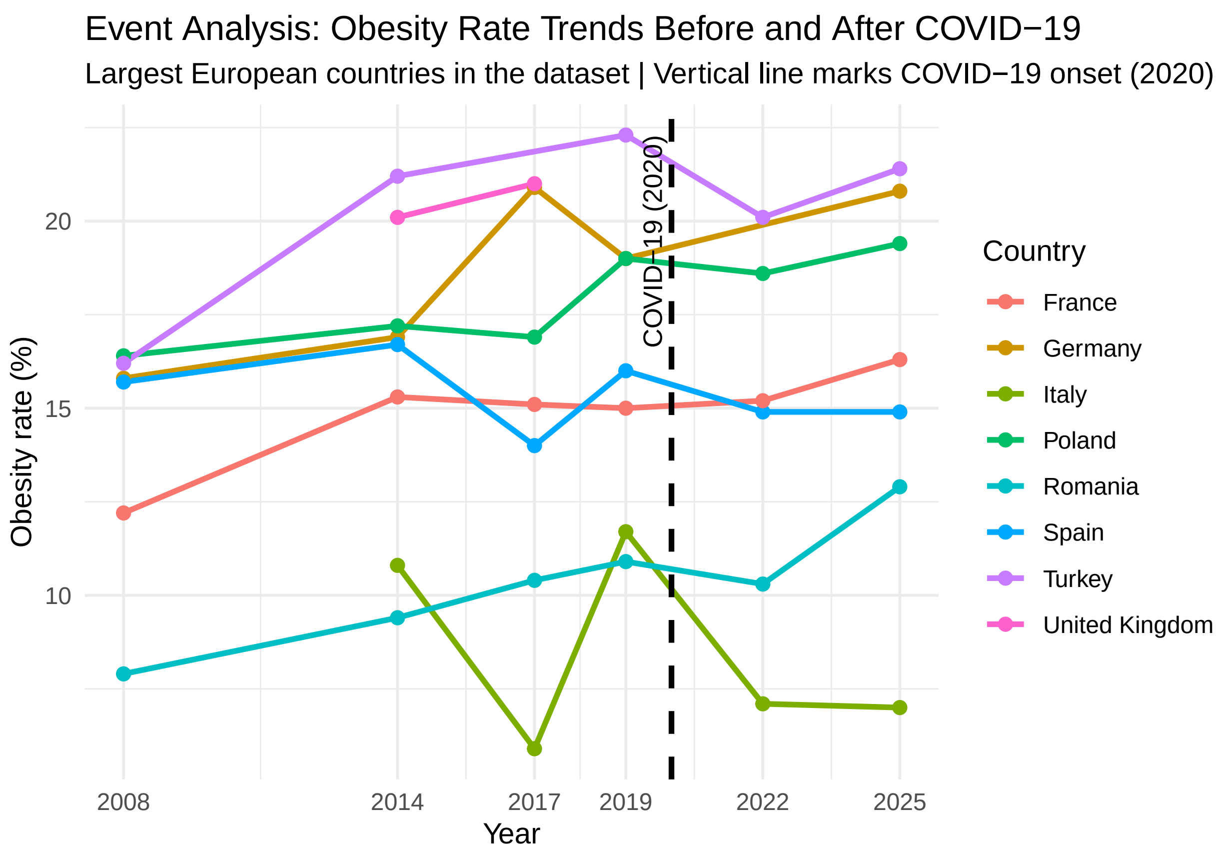
Obesity is most prevalent among the low educated (median around 19%), lower among the medium-educated (around 17%), and lowest among the high educated (around 12%), a difference of roughly 7.5 percentage points between the lowest and highest groups. Because this gradient holds across the full set of European countries in the sample rather than being driven by a few outliers, it confirms at the population level that the education-obesity link is a broad, consistent pattern.

3.6 Event analysis

In this section we treat the COVID-19 pandemic as an event and look at the obesity rates of the biggest European countries in the dataset.

```r
# Event: COVID-19 (2020)
# A temporal graph showing obesity rate trends over time for the largest
# European countries, with a clear vertical line marking the COVID-19 event.

# Reuse the same country selection as section 3.3
largest_european_countries_in_dataset <- c("TR", "DE", "UK", "FR", "IT", "ES", "PL", "RO")

selected_countries <- largest_european_countries_in_dataset[
  largest_european_countries_in_dataset %in% obesity_trend_clean$geo
]

event_trend_data <- obesity_trend_clean %>%
  filter(geo %in% selected_countries) %>%
```

```r
  mutate(
    country_name = case_when(
      geo == "TR" ~ "Turkey",
      geo == "DE" ~ "Germany",
      geo == "UK" ~ "United Kingdom",
      geo == "FR" ~ "France",
      geo == "IT" ~ "Italy",
      geo == "ES" ~ "Spain",
      geo == "PL" ~ "Poland",
      geo == "RO" ~ "Romania",
      TRUE ~ geo
    )
  )

ggplot(event_trend_data, aes(x = TIME_PERIOD, y = trend_value,
                             color = country_name, group = country_name)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  geom_vline(xintercept = 2020, linetype = "dashed",
             colour = "black", linewidth = 1) +
  annotate("text", x = 2020, y = max(event_trend_data$trend_value, na.rm = TRUE),
           label = "COVID-19 (2020)", angle = 90, vjust = -0.5, hjust = 1,
           size = 3.5, colour = "black") +
  scale_x_continuous(breaks = sort(unique(obesity_trend_clean$TIME_PERIOD))) +
  labs(
    title    = "Event Analysis: Obesity Rate Trends Before and After COVID-19",
    subtitle = "Largest European countries in the dataset | Vertical line marks COVID-19 onset (2020)",
    x        = "Year",
    y        = "Obesity rate (%)",
    color    = "Country"
  ) +
  theme_minimal()
```



This plot treats the COVID-19 pandemic as an event, with the dashed vertical line marking the start of the pandemic in 2020. This plot tracks the obesity rates for the largest European countries in the data set across the surrounding years. The reason only the largest European countries are included is because, if all EU countries would be included, the legend would be too cluttered to be useful. Contrary to the common expectations that lockdowns would push obesity upward after 2020, the trends after the event showed that obesity in most countries even decreased or stayed similar. This suggests that either the pandemic was not a uniform shock to measured obesity, or that self reported survey data does not fully capture what happened during these years. The analysis is purely descriptive and cannot establish that COVID-19 caused these changes, partly because survey waves are several years apart and do not allow us to observe a sharp before/after jump. A further limitation is that the obesity trend data carries no education breakdown for 2022, so while we can track the overall rate around the event we cannot test the question most central to our project; whether the pandemic widened or narrowed the gap between low- and high-educated groups.

Part 4 - Discussion

4.1 Discuss your findings

Using two Eurostat datasets on BMI, education, and obesity trends, we cleaned, filtered, and merged the data, then derived three variables: an education label, a low vs high education obesity gap, and the change in that gap between waves. The temporal analysis showed that obesity rates have generally increased across the largest European countries over the long run, though the pace and starting points differ considerably. Spatially, the education-obesity gap varies strongly across countries, with Croatia, Turkey and Luxembourg showing the largest gaps and Estonia showing a near zero or even reversed pattern. The sub population boxplots confirmed that lower education is consistently associated with higher obesity rates across all countries in the dataset. Finally, the event analysis treated the COVID-19 pandemic as a shock and compared overall obesity rates before (2019) and after (2022): contrary to the common expectation that the pandemic would raise obesity, most countries actually recorded a slight decrease, a counter intuitive result that raises questions about whether the self reported survey data fully captures what happened during this period. These findings carry clear limitations: self reported BMI likely understates obesity, the analysis is purely descriptive, the country level aggregates risk an ecological fallacy, and the absence of an education breakdown for 2022 means we cannot measure how the pandemic itself affected the gap. Even so, the harmonized Eurostat data and the consistent education gradient across the temporal, spatial, and sub population analyses still make the education-obesity link convincing.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/theodorkubi/ProgrammingForEcon

5.2 Reference list

European Commission. (2024). Health inequalities: Overweight and obesity. Knowledge4Policy. https://knowledge4policy.ec.europa.eu/health-promotion-knowledge-gateway/health-inequalities-5_en

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Eurostat. (n.d.-b). Obesity rate by body mass index (BMI) [Data set]. Retrieved June 7, 2026, from https://ec.europa.eu/eurostat/databrowser/view/sdg_02_10